OO-L-90a
September 14, 1962

SUPERSEDING
Fed. Spec. OO-L-90
April 18, 1956

FEDERAL SPECIFICATION

LATHE, BRAKE-DRUM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers floor mounted, electrically powered lathes, with equipment and accessories for truing brake drums of commercial and military series automotive vehicles.

1.1.1 *Federal specification coverage.* Federal specifications do not cover all sizes of lathes which are commercially available, but only those which are generally used by the Federal Government.

1.2 Classification.

1.2.1 *Sizes.* Brake-drum lathes covered by this specification shall be of the following sizes, as specified (see 6.2).

Size I.—Standard-duty.

Size II.—Heavy-duty.

Size III.—Extra-heavy-duty.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

W-C-596—Connector, Plug, Electrical; Connector, Receptacle, Electrical; Plate, Wall, Electrical.

CC-M-636—Motors, Fractional Horsepower (Alternating Current).

CC-M-641—Motor, Alternating-Current (Integral-Horsepower).

FF-B-171—Bearings, Ball, Annular (General Purpose).

FF-B-185—Bearings, Roller, Cylindrical, and Bearings, Roller, Self-Aligning.

FF-B-187—Bearing, Roller, Tapered.

TT-E-485—Enamel, Semi-Gloss, Rust-Inhibiting.

TT-E-489—Enamel, Alkyd, Gloss, for Exterior and Interior Surfaces.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Stand-

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ards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

JAN-P-100—Packaging and Packing for Overseas Shipment, General Specification.

MIL-C-104—Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.

MIL-P-116—Preservation, Methods of.

MIL-V-173—Varnish, Moisture- and Fungus-Resistant, for the Treatment of Communications, Electronic, and Associated Electrical Equipment.

MIL-T-704—Treatment and Painting of Material.

MIL-V-1137—Varnish, Electrical Insulating (for Electro-Motive Equipment).

MIL-T-7928—Terminals; Lug and Splice, Crimp Style, Copper, Insulated.

MIL-I-11748—Interference Reduction for Electrical and Electronic Equipment.

MIL-S-12514—Starter, Motor, Across-the-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100-HP, Manual and Magnetic.

MIL-I-16910—Interference Measurement, Radio, Methods and Limits; 14 Kilocycles to 1,000 Megacycles.

MIL-I-26600—Interference Control Requirements, Aeronautical Equipment.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-130—Identicifation Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Bureau of Standards Handbook:

H28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.).

American Standards Association Standards:

B5.17-1958—Markings for Identifying Grinding Wheels.

B6.1 -1932—Spur Gear Tooth Form.

B6.6 -1946—Gear Tolerances and Inspection.

B7.1 -1956—American Standard Safety Code for the Use, Care, and Protection of Abrasive Wheels.

Cl-1959—National Electric Code.

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York 16, New York.)

National Machine Tool Builder's Association Standards:

Machine Tool Electrical Standards.

(Application for copies should be addressed to the National Machine Tool Builder's Association, 10525 Carnegie Avenue, Cleveland 6, Ohio.)

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Tire and Rim Association:
Military Tire and Rim Data Book.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 2001 First National Tower, Akron 8, Ohio.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified, the contractor shall furnish complete brake-drum lathe samples for preproduction inspection in accordance with 4.2 (see 6.2). Unless otherwise specified, one sample shall be provided for this inspection (see 6.2). When specified, a preproduction test report shall be prepared and submitted to the contracting officer for approval (see 6.2). The preproduction sample shall be identical with the proposed production items. The approved sample and the preproduction report shall be used as the production standard.

3.2 Design. The brake-drum lathe shall include all the features, components, and accessories necessary for truing automotive brake drums as specified. The lathe shall be new, and shall be capable of meeting the performance requirements specified herein. The lathe shall be electric-motor-driven, with automatic and hand feeds. The equipment shall be convenient to operate from the front of the machine, and the operations such as feed selection, tool-bar positioning, grinder-quill positioning, movement of all slides and main spindle traverse be within easy reach of the operator. All parts subject to wear, breakage, or distortion shall be accessible for replacement, adjustment and repair. Technical data for tires and rims for military ground vehicles are available from the Tire and Rim Association's Military Tire and Rim Data Book.

3.3 Construction. Construction shall be free from any characteristics or defects that may render the lathe unsuitable or inefficient for the intended purpose.

3.3.1 *Castings and forgings.* All castings shall be free from blowholes, porosity, hard spots, shrinkage defects, or cracks. Forgings shall be free of scale, inclosures, cold shuts, and mismatching that may affect the structural strength of the forgings involved. Strength and other physical and chemical properties of the castings and forgings shall be adequate to meet the performance requirements specified herein. Castings and forgings shall not be subjected to any reclaiming process without the authorization of the procuring activity.

3.3.2 *Welding, brazing, and soldering.* Welding, brazing, and soldering shall be performed in accordance with commercial practice.

3.3.3 *Fastening devices.* All screws, pins, bolts, and similar fastening devices shall be installed with means for preventing loss of tightness and adjustment. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 *Standard parts.* Military standard parts or commercial standard parts shall be used provided they conform to the requirements of this specification, except as otherwise specified herein.

3.3.5 *Electrical connections.* Electrical connections of conductors and terminal parts shall be by the pressure or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured and soldered to insure electrical continuity. Flux for soldering shall be a resin base. Crimp-style terminals shall conform to the requirements of MIL-T-7928.

3.3.6 *Threads.* All threaded parts shall comply with the requirements of National Bureau of Standards Handbook H28. Unified screw threads shall be utilized in lieu of American National screw threads and non-unified American Standard screw threads.

3.3.7 *Gears.* All gears and pinions shall be fabricated in the English system and shall conform to the requirements of American Standards Association Standard (ASA)

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B6.6. If tolerances other than those specified are required to manufacture gears or pinions, such tolerances shall assure accuracy of operation of the lathe equal to or exceeding that specified herein. Spur-gear tooth form shall conform to the requirements of ASA B6.1.

3.3.8 *Bearings.* Bearings shall be sleeve type or antifriction type. When other than tapered roller bearings are used, suitable thrust bearings shall be provided. Antifriction bearings when furnished shall be interchangeable with, or be in accordance with, the requirements of FF-B-171, FF-B-185, or FF-B-187, as applicable. Sleeve-type bearings shall be bronze or babbitt. All bearings shall be protected from dust and chips.

3.3.9 *Lubrication.* When necessary to reduce wear and maintain efficiency, means shall be provided for lubrication except permanently lubricated sealed-for-life mechanisms.

3.3.10 *Safety devices* Each machine shall be furnished with safety devices. Parts which are hazardous to the operator shall be guarded. Protection against electric shock shall be provided. When specified, special safety devices shall be provided as required by the procuring activity (see 6.2).

3.3.11 *Interchangeability.* All replaceable parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model can be interchanged, replaced, or adjusted without requiring modification. All replaceable parts shall be permanently and legibly marked with the manufacturer's part number in accordance with MIL-STD-130 except commonly known commercial items which present no identification problems such as nuts, bolts, and washers.

3.3.12 *Surfaces.* External surfaces shall be cleaned and smoothed, and all edges shall be rounded of beveled. All bearing surfaces between matching parts shall be finished by machining, grinding, or scraping to necessary tolerances to provide the required accuracy. All castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux, and other harmful materials.

3.3.13 *Materials.* Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subjected to all provisions of this specification.

3.3.13.1 *Corrosion resistance.* All exposed metallic materials shall be inherently corrosion resistant or treated to be corrosion resistant.

3.4 Performance and product characteristics. The lathe shall be a power-driven machine tool capable of truing commercial or military series vehicles brake drums of the sizes specified. The sizes II and III lathes shall permit turning and grinding the brake drums while mounted on wheels, including dual wheels, without deflating or removing the tires from wheels. When specified, the sizes II and III lathes shall be capable of truing brake drums without the use of a bearing or outboard support (see 6.2). Unless otherwise specified, means shall be provided for attaching a grinding attachment to the lathe (see 6.2).

3.4.1 *Capacity.* The lathe shall be furnished in the following sizes and embody the physical characteristics specified in table I:

3.4.1.1 *Size I, standard duty.* The size I, standard-duty lathe shall be capable of truing brake drums within the sizes specified (see table I). The lathe shall provide for truing the brake drums of the military vehicles shown in table II. Unless otherwise specified, the size I lathe shall not require the truing of brake drums while mounted on wheels with tires (see 6.2).

3.4.1.2 *Size II, heavy-duty.* The size II, heavy-duty lathe shall be capable of truing vehicle brake drums within the size range specified (see table I). The lathe shall provide for truing the brake drums for the military vehicles shown in table II.

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TABLE I.—*Physical characteristics*

	Size I	Size II	Size III
Drum capacity range (dia) inches (Min)	7-1/2 to 18	8.9 to 25	9 to 36
Drum Cutting depth (face), inches (Min)	3/4 to 7½	10	15
Speeds, number of changes (min)	3	3	3
Speeds, cutting, range, r.p.m. ± 10	50 to 110	30 to 100	30 to 100
Swing, inches (min)	36	60	72
Horsepower, drive motor (min)	1/2	1	2
Horsepower, grinding motor, when applicable, (min)	1/4	1/2	1/2
Feed, inches per revolution, range (min)	.005-.020	.005-.016	.0025-.030
Number of feeds (min)	2	2	3
Weight, pounds (max)	500	—	—

TABLE II.—*Brake drum characteristics, military vehicles*

SNL	*MODEL*	*HUB NO*	*DIA. X DEP.*	*ORD ENG STD*	*CONE NO*	*LOC*	*ORD ENG STD*	*CUP NO*	*LOC*
G842	XM455 XM456	8719915	14.9 x 3.3	CAS X 5.1	712869		CAS X 6.1	712868	
G751	M119 M118 M118A1 M164E1	8343586	16.4 x 6	CAS X 7.1 CAS X 6	706691 712120	IN OUT	CAS X 7.1 CAS X 6	712286 712119	IN OUT
G815	M349	8720968	16.5 x 6.7	CAS X 5 CAS X 4	706683 706671	IN OUT	CAS X 6 CAS X 4	705143 705116	IN OUT
G800	M197	8327340	16.4 x 6	CAS X 7.1 CAS X 4	706691 706671	IN OUT	CAS X 7.1 CAS X 4	712286 705116	IN OUT
	M197A1	8747994		CAS X 7.1 CAS X 4	706691 706671	IN OUT	CAS X 7.1 CAS X 4	712286 705116	IN OUT
	M198	7409395		CAS X 7.1	706691		CAS X 7.1	705155	OUT
G816	M345	8710723	16.4 x 6	CAS X 7.1	706691	IN OUT	CAS X 7.1 CAS X 7.1	712286 705155	IN OUT
G750	M126 M127 M126A1 M127A1 M128A1	7409395 8333869	16.4 x 6	CAS X 7.1 CAS X 7.1 CAS X 7.1	706691 706691 712286	 IN OUT	CAS X 7.1 CAS X 7.1 CAS X 7.1 CAS X 7.1	712286 705155 712286 705155	IN OUT IN OUT
G755	M131	7409395	16.4 x 6	CAS X 7.1	706691		CAS X 7.1 CAS X 7.1	712286 705155	IN OUT
	M131A1 M131A2	8333869		CAS X 7.1	706691		CAS X 7.1 CAS X 7.1	712286 705155	IN OUT
G802	M269 M270	7979324	16.4 x 6	CAS X 7.1	706691		CAS X 7.1 CAS X 7.1	712286 705155	IN OUT
G797	M172	8379664		CAS X 6 CAS X 5.1	706704 707081	IN OUT	CAS X 7 CAS X 6	705187 705167	IN OUT
	M172A1	8787067RT 8687068LH	12.2 x 6.6	CAS X 7 CAS X 5.1	706704 707081	IN OUT	CAS X 7 CAS X 6	705187 705167	IN OUT
G811	M199	8720685	16.4 x 7.5	CAS X 6.1 CAS X 7	706707 706704	IN OUT	CAS X 7 CAS X 7	705194 705187	IN OUT
G160	M15A1 M15A2	7348199	16 x 6	CAS X 7 CAS X 5.1	706704 706784	IN OUT	CAS X 7 CAS X 5.1	705187 705341	IN OUT

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TABLE II.—*Brake drum characteristics, military vehicles (cont'd.)*

SNL	HUB ORD NO	CUP MFG'S NO*	LOC	CONE MFG'S NO*	LOC	BRAKE DRUM DIA. IN.	BRAKE DRUM DEPTH IN.
G740 & G758	7375142	18520	INF	18590	INF	8.948	2 17/64
	7375144		OUTF		OUTF		
Rear wheel hubs are taper bored to fit tapered shaft							
G741	8376840	28622	INF	28682	INF	14.155	2 11/32
			INR		INR		
		2924	OUTF	2984	OUTF		
			OUTR		OUTR		
G742	8357986	3920E	INF	3994	INF	14.995	3 11/16
			INR		INR		
		3920E	OUTF	392	OUTF		
			OUTR		OUTR		
G749	GM2280042	39520	INF	39590	INF	14.995	3 11/32
			INR		INR		
		39520	OUTF	39590	OUTF		
			OUTR		OUTR		
G744	7979324	592A	INFR	593A	INFR	16.495	6 1/16
		592A	OUTRF	596	OUTRF		
G792	M34KM46	5735	INF	5760	INF	17.25	6 15/32
	M12QJ451	522	OUTF	528A	OUTF		
		64700	INR	64450	INR		
		854	OUTR	861	OUTR		

IN—inner
OUT—outer
F—front
R—rear
*Timken

3.4.1.3 *Size III, extra-heavy-duty* The size III, extra-heavy-duty lathe shall be capable of truing brake drums within the size range specified (see table I).

3.4.2 *Accuracy.* The brake-drum lathe shall be capable of truing brake drums in accordance with the following tolerances:

(*a*) Concentricity—within 0.002-inch total indicator reading, with workhead spindle.
(*b*) Taper—0.001-inch per inch maximum.
(*c*) Runout depth of mounted drum—0.004-inch maximum total indicator reading.
(*d*) Final surface—50 microinches, maximum (root-mean-square averaging method).

3.5 **Detail of components.** The lathe shall consist of the following principal component parts:

(*a*) Base.
(*b*) Carriage or sliding spindle.
(*c*) Spindle and arbors.
(*d*) Cross-slide.
(*e*) Tool bar.
(*f*) Electrical equipment.
(*g*) Case (when specified).
(*h*) Grinding attachment (when specified).
(*i*) Crane attachment (when specified).
(*j*) Accessories.
(*k*) Spare parts and tools.

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3.5.1 *Base.* The base shall be of cast iron or steel fabrication. Beds shall be seasoned or stress relieved. Steel-weldment structures shall be stress relieved. The base shall be of sufficient strength to insure rigidity and reduce vibration.

3.5.1.1 *Ways.* All ways shall be finished to provide an accurate, uniform bearing surface for the matching ways. All V-way and flat-way construction shall be scraped or ground to maintain the required accuracy.

3.5.2 *Carriage.* The carriage, when the lathe is not the sliding spindle type, shall be made of cast iron and shall slide on dovetail ways with means provided for adjustment to compensate for wear.

3.5.2.1 *Feed drive.* The feed drive shall be totally enclosed; feed changes shall be easily adjustable. A handwheel feed or clutch for engaging the automatic feed shall be provided. If a feed screw is provided, it shall operate in a bronze nut. The nut shall be detachable for easy replacement.

3.5.2.2 *Automatic stops.* An automatic-stop device shall be provided which will shut-off the feed at a predetermined point, preventing damage to the equipment for both grinding and turning.

3.5.3 *Spindles and arbors.* Spindles and arbors shall be high-grade alloy steel, properly heat treated and ground on all aligning surfaces. Spindles and arbors shall be of a size to insure the accuracy required of this type of lathe and so mounted as to permit turning brake drums of the size specified. The arbors shall be mounted in the spindle by means of a suitable taper with a positive lock and release system which will prevent damage to spindle or arbor. The length of the arbor shall be adequate to mount brake drums of all vehicles within the sizes specified.

3.5.3.1 *Spindle drive.* Spindle drive shall be through V-belts, chains, gears, or combinations. Spindle speeds shall be variable through use of stepped pulleys and infinitely variable drive mechanism or quick-change-speed gears. A minimum of three spindle speed ranges shall be provided. Gear drives shall be totally enclosed. Easily adjustable tensioning devices and guards shall be provided where belt or chain drive is employed. All necessary motor mounting brackets, pulleys, belts, chains, sprockets, gears, countershafts, and other parts required to put the lathe in successful operation shall be furnished.

3.5.4 *Cross-slide.* The manual cross-slide shall be made of cast iron and shall be equipped with a feed dial graduated in increments to indicate 0.001-inch movements of the cross-slide. The cross-slide shall have provisions for locking in any position and adjustment to compensate for wear.

3.5.5 *Tool bar.* The lathe tool bar shall be made from alloy steel and shall be capable of being clamped rigidly to the cross-slide. The tool bar shall be designed so that it can be rotated 180 degrees in the clamp for turning drums with external braking surfaces.

3.5.6 *Electrical equipment.*

3.5.6.1 *Motors.* The motors shall conform to the requirements of CC-M-636 and CC-M-641 and table I, as applicable. The operating voltage, current, frequency and phase shall be as specified herein. Mounting shall be in accordance with the Electrical Standards of the National Machine Tool Builder's Association (NMTBA). Motors shall be protected from dirt and grit and shall be accessible for lubrication and maintenance and shall be equipped with sealed-for-life, antifriction bearings. A start-stop switch with overload protection shall be included and wired.

3.5.6.1.1 *Size I and size II.* Unless otherwise specified (see 6.2), the size I and size II lathes shall be furnished with a 115-volt, 60-cycle, single-phase capacitor motor, 50-degree Centigrade ambient temperature use, open, continuous-duty drive motor, and a

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grinding motor of similar characteristics when a grinding attachment is specified (see 3.5.8).

3.5.6.1.2 *Size III.* Unless otherwise specified (see 6.2), the size III lathe shall be furnished with a 220-volt, 60-cycle, 3-phase, open, continuous-duty drive motor, and a grinding motor of similar characteristics when a grinding attachment is specified (see 3.5.8). If a 3/4 horsepower or more induction motor is used, it shall be equipped with a starter that shall conform to the requirements of MIL-S-12514, type I, class 2.

3.5.6.2 *Electrical components.* Electrical components of the lathe including starters, relays, switches, wiring, and their inclosures shall conform to and shall be located in accordance with the NMTBA Electrical Standards, except as specified herein. No component part shall be used where the requirements exceed the manufacturer's rating for the part.

3.5.6.3 *Light fixture.* The lathe shall be provided with a work light located in a position which will not interfere with controls, yet will afford sufficient illumination necessary for the turning or grinding operations.

3.5.6.4 *Extension cord.* When specified, the sizes I and II lathes shall be furnished with not less than 8 feet of 3-conductor flexible electrical cord in accordance with the requirements of ASA C1, National Electric Code, type SO, wire size as specified therein for current requirements (see 6.2). The cord shall terminate in an appropriate 3-pole grounding plug conforming to the requirements of W-C-596, style D. A mating three to two parallel prong adapter with a grounding lead shall be furnished to adapt the plug for use with a two-prong outlet.

3.5.7 *Case.* When specified, a suitable sturdy case of sheet metal or wood shall be furnished to contain grinding attachment, gages, and all accessories for transportation of the equipment without damage (see 6.2).

3.5.8 *Grinding attachment.* When specified (see 6.2), a grinding attachment shall be furnished with each lathe with six grinding wheels suitable for grinding cast-iron and hardened-steel brake drums. A suitable guard, an abrasive-wheel dresser having a mounted octahedron of 1/4-carat minimum size, feed mechanism, brackets, and necessary equipment to make a complete unit shall be included. The grinding attachment shall be capable of driving the abrasive wheel at a working surface speed of 3600 to 7000 feet per minute.

3.5.8.1 *Grinding wheels.* The grinding wheels shall be marked, handled, mounted, and used in accordance with ASA B5.17 and ASA B7.1.

3.5.9 *Crane attachment.* When specified (see 6.2), a crane attachment (less chain falls) shall be provided for the sizes II and III lathes and that attachment shall be capable of being attached to the lathe in such a manner as to form an integral unit. The crane shall be of sufficient strength to handle all the drums with tires and wheels attached, within the capacities specified.

3.5.10 *Accessories.* The accessories for each lathe including all arbors, cones, spacers, faceplates, floating drum adapters, and other devices required in truing brake drums of commercial and military series of vehicles specified by the procuring activity shall be furnished (see 6.2). A silencer shall be provided to prevent chatter while turning or grinding brake drums.

3.5.11 *Spare parts and tools.* The following spare parts and tools shall be furnished with each lathe:

1 Complete set of extra belts, when used.
1 Complete set of tools (other than common) necessary for operation of the equipment.
2 Tool bits, high-speed steel for cast iron.
2 Tool bits, carbide for steel and cast iron.

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3.6 Suppression of radio interference.

3.6.1 *Army.* Unless otherwise specified (see 6.2), the equipment shall meet the limits of MIL-I-11748 for class IIIc over the frequency range of .15 to 1000 megacycles. Application of suppression components and techniques shall conform to the specification. If the contractor considers it desirable to modify the equipment proper in order to facilitate conformance with the specified limits, he shall obtain approval from the contracting officer before proceeding with the modification (see 6.3).

3.6.2 *Navy.* When specified, the electrical equipment covered by this specification shall comply with MIL-I-16910 relative to radiated and conducted radio interference (see 6.2 and 6.3).

3.6.3 *Air Force.* When specified (see 6.2), the electrical equipment covered by this specification shall conform to the requirements of MIL-I-26600, when motors capable of producing radio interference are used. Induction or repulsion induction type motors without communication devices or those using brush lifting devices allowing brushes to make contact only during the starting cycle are exempt from radio-interference suppression treatment or test requirements (see 6.2 and 6.3).

3.7 Fungus control. When specified (see 6.2), circuit elements which are not inherently fungus resistant such as cable and wire (except plastic, rubber, and bitumen coated) shall be coated with a fungicidal varnish conforming to the type I requirements of MIL-V-173. The windings of the electrical motor shall be given two impregnating coats of varnish conforming to type M, grade BB or CB and one sealer coat conforming to type M, grade CA of the requirements of MIL-V-1137. The coats shall be applied by a method which shall completely fill all interstices in the coils and preclude the possibility of entrapping air or moisture. The following electrical components shall not be treated:

(a) Switches.

(b) Fuses.

(c) Contacts.

3.8 Painting. Unless otherwise specified, painting and finishing of the lathe and case shall be as follows (see 6.2).

3.8.1 *Preparation of surfaces.* All surfaces to be painted shall be cleaned of all grease, oil, scale, rust, dirt, casting sand, slag, and other extraneous matter and shall be dry.

3.8.2 *Application.* Application shall be in accordance with the requirements of MIL-T-704, except that the lathe shall be finished with one coat of primer conforming to the requirements of TT-E-485 followed by not less than one coat of enamel conforming to the requirements of TT-E-489. Color shall be No. 16187 (gray) as shown in Fed. Std. No. 595. The interior of gear cases or similar compartments and cast-in reservoirs shall be protected by a sealer that will withstand immersion in lubricating oil, hydraulic fluids, and cutting compounds at the usual working temperatures and atmospheric conditions.

3.9 Identification of product. The lathe shall be marked for identification in accordance with MIL-STD-130.

3.10 Handbook of instructions. Unless otherwise specified, the contractor shall furnish two copies of a handbook of instructions with each lathe supplied under this specification (see 6.2). The handbook shall contain necessary operating instructions including list of accessories specified for each vehicle brake drum, drawings, illustration, schematics, parts list, manufacturer's part numbers, lubricating instructions, assembly and disassembly instructions and safety precautions, to insure proper operation and maintenance of the lathe. Additional copies of the instruction handbook and the mailing address to which additional copies shall be sent will be specified by the procuring activity (see 6.2). Unless otherwise specified, the handbook shall contain the manufacturers part numbers for the accessories for the brake drums listed in table II for sizes I and II lathes see (6.2).

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3.10.1 *Handbook approval.* When specified, a preliminary copy of the instruction handbook shall be submitted to the contracting officer for approval prior to publication and delivery (see 6.2).

3.11 **Workmanship.** Standard of workmanship shall be such that equipment specified herein shall comply with the requirements in section 3, and final assembly shall comply with tests in section 4. Workmanship shall be of such quality as to assure a product free of defects, such as bulges, warpage, dirt, fins, sprues, scale, flux, burrs, scratches, cuts, pits, sharp edges, rust, cold solder joints, defective connection, loose, damaged, or missing parts.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Preproduction inspection.** Unless otherwise specified (see 6.2), preproduction inspection shall consist of all the test specified in 4.4 and 4.6 and examination in accordance with 4.3.2 except that sampling in accordance with MIL-STD-105 shall not apply. Preproduction samples shall be as specified in 3.1. Preproduction inspection shall be preformed by the contractor. Failure of preproduction sample to pass preproduction inspection shall be cause for rejection.

4.3 **Acceptance inspection.** Acceptance inspection shall consist of the examination specified in 4.3.2 and the test specified in 4.3.3.

4.3.1 *Acceptance lot.* An acceptance lot shall consist of all the same size lathes from one manufacturer submitted for acceptance at one time.

4.3.2 *Acceptance examination.* The sample lathes from each lot shall be selected in accordance with MIL-STD-105 at inspection level II and examined using the following classification of defects and acceptable quality level (AQL). Any lathe found defective shall be rejected, and if the number of defective lathes exceeds the AQL limits specified, the entire lot represented by the samples shall be rejected.

4.3.2.1 *Classification of defects.*

Major Defects
(AQL shall be 2.5 percent defective)

101. Component parts not new; evidence of reclaimed, used, or salvaged components (see 3.2).
102. Parts not accessible for repair (see 3.2).
103. Equipment not convenient to operate (see 3.2).
104. Evidence that the equipment will not operate (see 3.3).
105. Defective castings and forgings affecting operation (see 3.3.1).
106. Defective welding, brazing, and soldering affecting operation (see 3.3.2).
107. Standard parts not as specified (see 3.3.4).
108. Lubrication not as specified (see 3.3.9).
109. Safety devices not as specified (see 3.3.10).
110. Interchangeability and marking not as specified (see 3.3.11).
111. Brake drum size range capacities not as specified (see table I).

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112. Characteristics not as specified in table I (see 3.4 thru 3.4.1.3).
113. Base, ways not as specified (see 3.5.1 and 3.5.1.1).
114. Carriage not as specified (see 3.5.2).
115. Feed drive not as specified (see 3.5.2.1).
116. Automatic stop not as specified (see 3.5.2.2).
117. Spindles and arbors not as specified (see 3.5.3).
118. Spindle drive not as specified (see 3.5.3.1).
119. Cross-slide not as specified (see 3.5.4).
120. Tool bar not as specified (see 3.5.5).
121. Electrical equipment not as specified (see 3.5.6 thru 3.5.6.4).
122. Grinding attachment not as specified (see 3.5.8).
123. Grinding wheels not as specified (see 3.5.8.1).
124. Accessories not as specified (see 3.5.10).
125. Spare parts not as specified (see 3.5.11).

(AQL shall be 4.0 percent defective)

126. Electrical connections not as specified (see 3.3.5).
127. Threads not as specified (see 3.3.6).
128. Gears not as specified (see 3.3.7).
129. Bearings not as specified (see 3.3.8).
130. Case missing or not as specified (see 3.5.7).
131. Identification of product not as specified (see 3.9).

Minor Defects
(AQL shall be 6.5 percent defective)

201. Surfaces not as specified (see 3.3.12).
202. Painting not as specified (see 3.8).
203. Handbook of instructions missing or incomplete (see 3.10 and 3.10.1).
204. Workmanship not as specified (see 3.11).

4.3.3 *Acceptance tests.* From each lot which has passed the acceptance examination specified in 4.3.2, samples for acceptance tests shall be selected in accordance with inspection level L7 of MIL-STD-105. The samples selected shall be subjected to the performance test (see 4.4.1). Acceptance or rejection shall be based on an AQL of 1 percent defective.

4.4 Test methods.

4.4.1 *Performance test.* One used, reclaimable brake drum of the largest diameter available, and one of the smallest diameter available, within the applicable size range specified for the size of lathe to be tested (see table I), shall be set up and trued as a typical operation. The concentricity, taper, runout, and surface roughness value shall be within the limits specified (see 3.4.2).

4.4.1.1 *Size I.* Unless otherwise specified, the brake drums for testing with size I lathe as specified in 4.4.1, are not required to be mounted on a wheel assembly (complete with tires) (see 3.4.1.1).

4.4.1.2 *Sizes II and III.* Unless otherwise specified, the brake drums for testing with sizes II and III lathes as specified in 4.4.1, are required to be mounted on wheel assembly (complete with tires) (see 6.2).

4.4.2 *Spindle speeds.* The lathe shall be tested with a tachometer to determine compliance with spindle speed range and number of speeds required. Failure to comply with minimum speed range and the minimum speed changes specified shall be cause for rejection (see table I).

4.5 Radio interference suppression test.

4.5.1 *Army.* The preproduction sample shall be tested as described in MIL-I-11748, class IIIc, over the frequency range of .15 to 1000 megacycles, by authorized representatives of the contracting officer. Unless otherwise specified, the preproduction sample will be tested by an Army Testing agency

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using Government test equipment (see 6.2). The testing agency will furnish the contracting officer a report on the model tested, including details of the suppression system applicable for production units and recommendations for quality control of production. Upon approval by the contracting officer, and provided all other requirements of this specification are met, the contractor shall examine each production unit to determine that it conforms to the report of the contracting officer and the approval suppression system of the preproduction sample.

4.5.2 *Navy and Air Force.* The preproduction sample shall be tested by the contractor or an approved testing laboratory, at the contractor's expense, for radio frequency radiation in accordance with MIL-I-16910 for Navy procurement, and MIL-I-26600 for Air Force procurement.

4.6 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the lathe shall be examined and tested to determine conformance with the requirements of section 5 of this specification, and as required by the specifications inspection lots and sampling shall be in accordance with MIL-STD-105, and an AQL of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

(Civil agencies. Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment).

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Disassembly.* The lathe may be disassembled to effect savings in cubage provided no special tools are required for reassembly.

5.1.1.2 *Cleaning.* The lathe shall be cleaned in accordance with C-1 process of MIL-P-116.

5.1.1.3 *Drying.* The lathe shall be dried in accordance with procedures D-1 and D-4 or a combination of both as stated in MIL-P-116.

5.1.1.4 *Methods.* Methods of preservation and packaging materials referenced by symbols herein shall conform to methods indicated by corresponding symbols in MIL-P-116.

5.1.1.5 *Unit package.* The basic unit of the lathe shall be packaged method IIa in accordance with MIL-P-116.

5.1.1.6 *Preservation.* The revolving and sliding spindle, micrometer control with dial, tool holder and cutter, and all other highly polished bare metal surfaces including spare parts and tool shall be coated with P-10 type preservative. When specified (see 6.2), the painted crane attachment will be supplied and packaged method III. The trolley shall be packaged method I with P-10 type preservative applied to the bearing areas. Procedure for preservation application of all items shall be in accordance with MIL-P-116.

5.1.1.7 *Consolidated packaging.* The extension cord, publications, and extra set of drive belts shall be packaged individually IC-1 in accordance with MIL-P-116. These items with other packaged component spare parts and tools shall be placed in the case provided for these items and secured within the exterior container.

5.1.2 *Level C.* The lathe with accessories, component parts, tools, and publications shall be cleaned, dried, preserved, and packaged in accordance with the supplier's commercial practice. Protection shall be sufficient to prevent damage or deterioration during shipment to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 *Level A.* Each lathe packaged method IIa shall be packed in a sheathed crate conforming to MIL-C-104, type II, class 1, style A. The packed tools and accessories in the case shall be arranged within the shipping container in such a manner as to afford complete protection from mechanical damage to the method II barrier of basic unit. Blocking, bracing and anchoring shall be in accordance with MIL-C-104 and JAN-P-100. Closure and strapping shall be in accordance with the crate specification. When specified (see 6.2), the crane attachment for size II heavy duty and size III extra heavy duty lathe shall be packed separately in a nailed wood exterior container conforming to PPP-B-621. Blocking, bracing and anchoring shall be in accordance with JAN-P-100.

5.2.2 *Level B.* Each lathe packaged method IIa shall be packed in a sheathed crate conforming to MIL-C-104, type II, class 1, style A. The packed tools and accessories in the case shall be arranged within the shipping container in such a manner as to afford complete protection from mechanical damage to the method II barrier of basic unit. Blocking, bracing and anchoring shall be sufficient to afford adequate protection against damage. When specified (see 6.2), the crane attachment for size II heavy duty and size III extra heavy duty lathe shall be packed separately in a nailed wood exterior container conforming to PPP-B-621. Blocking, bracing, and anchoring shall be sufficient to afford adequate protection against damage.

5.2.3 *Level C.* Each complete lathe with tool box and accessories shall be packed sufficiently for shipment to insure arrival at destination in satisfactory condition and acceptable to the carrier at lowest rates.

5.3 **Marking for shipment.**

5.3.1 *Military agencies.* Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 *Civil agencies.* In addition to any special marking specified in the contract or order, each unit and intermediate package and shipping container shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 **Intended use.** The brake-drum lathes covered by this specification are intended for use in truing brake drums of the commercial and military type.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Preproduction inspection when required, sample quantity, test report, if different (see 3.1).
(d) Safety devices if different (see 3.3.10).
(e) When sizes II and III lathes capable of turning brake drums without outboard support are required (see 3.4).
(f) When provisions for attaching a grinding attachment is not required (see 3.4).
(g) When size I lathe capable of truing brake drum with wheel and tire mounted is required (see 3.4.1.1 and 4.4.1.1).
(h) Voltage and current, if different (see 3.5.6.1 and 3.5.6.1.2).
(i) Cord, when required (see 3.5.6.4).
(j) Case, when required (see 3.5.7).
(k) Grinding attachment, when required (see 3.5.8).
(l) Crane attachment, when required (see 3.5.9).
(m) Specific vehicles for which accessories are to be furnished (see 3.5.10).
(n) When radio interference suppression is required (see 3.6.2 and 3.6.3).
(o) Fungus control, when required (see 3.7).

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(p) Painting, if different (see 3.8).

(q) Suppression of radio interference, if different (see 3.8.1).

(r) Handbook of instructions if different, and address where additional copies are to be sent (see 3.10).

(s) Preliminary handbook approval, when required (see 3.10.1).

(t) Preproduction inspection if different (see 4.2).

(u) Responsibility for preproduction inspection, acceptance inspection, and radio interference suppression if different (see 4.2 and 4.5.1).

(v) When brake drums mounted on wheel assembly is not required for sizes II and III lathe testing (see 4.4.1.2).

(w) Levels of preservation, packaging, and packing required (see 5.1 and 5.2).

(x) When crane attachment is to be packed separately (see 5.2.1 and 5.2.2).

6.3 Radio interference. When applicable, the contracting officer should arrange for investigation and test of the suppression components of the equipment by the Navy, Air Force, or Signal Corps in accordance with Section 6 of Military Specification MIL-I-16910, MIL-I-26600 Air Force, or MIL-I-11748 Army.

6.4 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Machines, not otherwise indexed by name. (Specify set-up or knocked-down.)
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Machines, not otherwise indexed. (Specify set-up or knocked-down.)
Motor volume minimum weight 24,000 pounds.

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